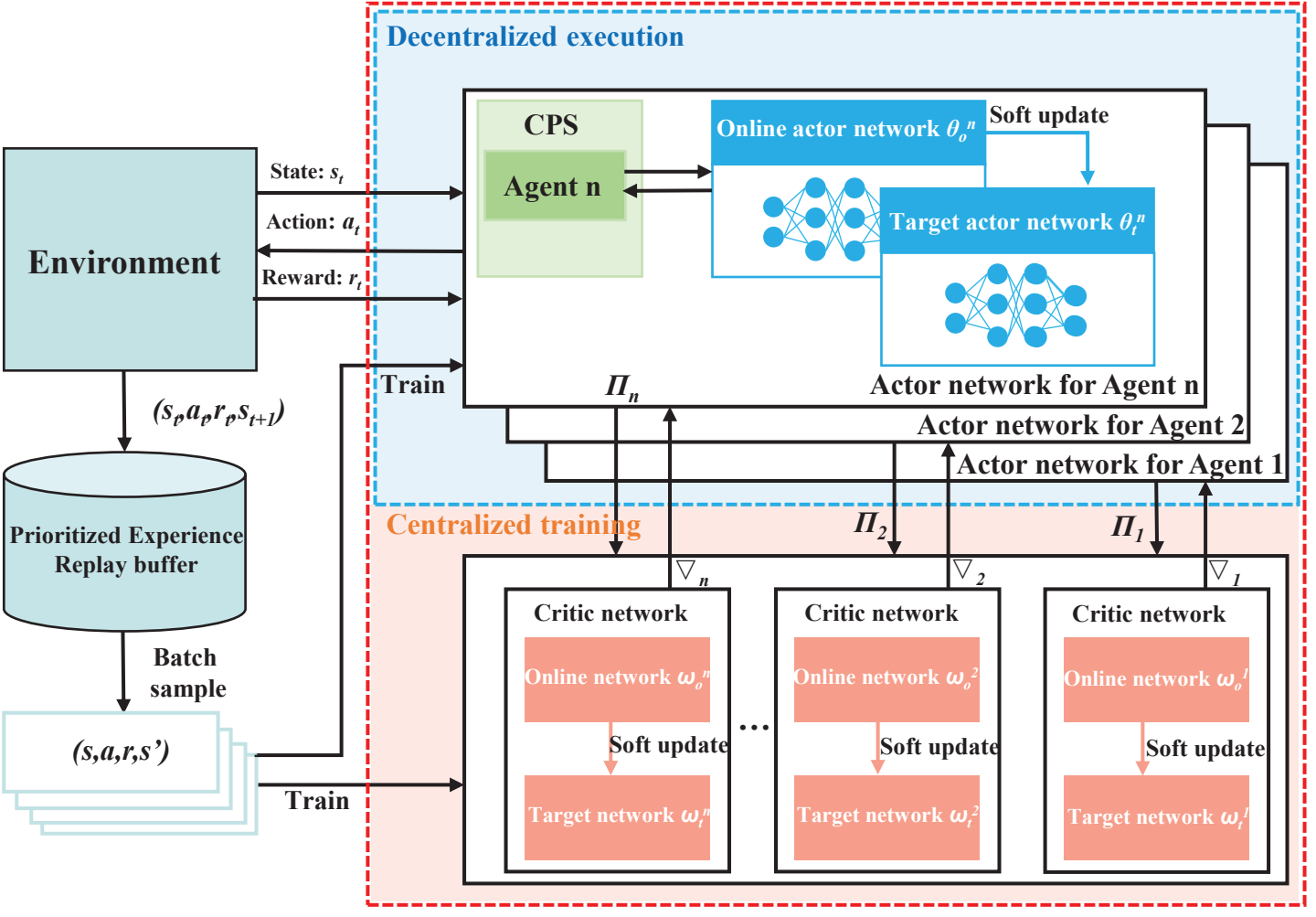

Decentralized execution
CPS
Agent n
Online actor network θ_o^n
Soft update
Target actor network θ_t^n
Environment
State: s_t
Action: a_t
Reward: r_t
Train
Π_n
Actor network for Agent n
Actor network for Agent 2
Actor network for Agent 1
(s_t, a_t, r_t, s_{t+1})
Prioritized Experience Replay buffer
Batch sample
(s,a,r,s')
Train
Centralized training
Π_2
Π_1
∇_n
∇_2
∇_1
Critic network
Online network ω_o^n
Soft update
Target network ω_t^n
...
Critic network
Online network ω_o^2
Soft update
Target network ω_t^2
Critic network
Online network ω_o^1
Soft update
Target network ω_t^1